

Turning PYTHIA jets into JEWEL jets with flow matching

An unpaired translation pilot on clean calorimeter jet images

October 2026

seed 0, 2 GPU hours per model, 20k held-out jets per sample; full log: `FLOW_NOTES.md`, results:
`docs/flow/translation`

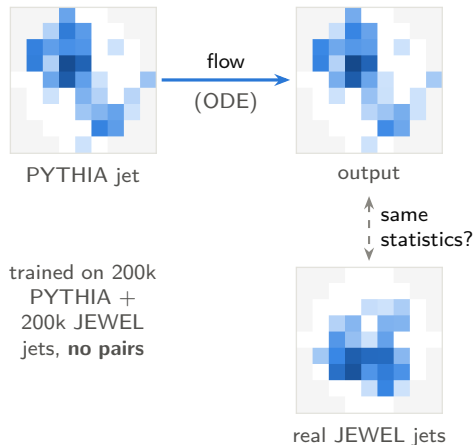
The question

Can a model trained on **unpaired** samples map a clean PYTHIA (vacuum) jet to a JEWEL-like (medium) jet?

Two things must hold:

1. the outputs look like real JEWEL jets, statistically;
2. each output still depends on its own input jet.

There is no true JEWEL partner for a PYTHIA jet: any unpaired map is one of many with the same distributions. So this tests distributions and dependence, not a per-jet medium modification.



Setup

Data: clean jet images, no background

- PYTHIA 2.6M events, JEWEL 870k events
- the leading $R = 0.4$ jet (≥ 10 GeV): 53 towers in GeV on a 16×16 grid
- per sample: 200k training jets, 20k held-out test jets, split by event

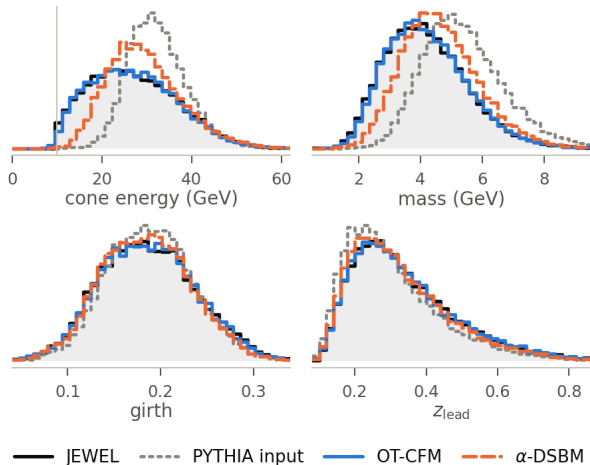
Compared with

- α -**DSBM**: learns its own, random pairing (same network and budget)
- the PYTHIA jet unchanged; a second JEWEL sample (best possible)

Model: **OT-CFM**, optimal-transport flow matching

- pair each batch of 256 PYTHIA and 256 JEWEL jets by optimal transport (least change in log energy)
- learn the velocity along straight lines between pairs (UVCAN-S network, 32M parameters; 2 hours on one A6000)
- translate a jet by solving the flow ODE: 128 network calls, 6.6 ms per jet

The output distributions match JEWEL

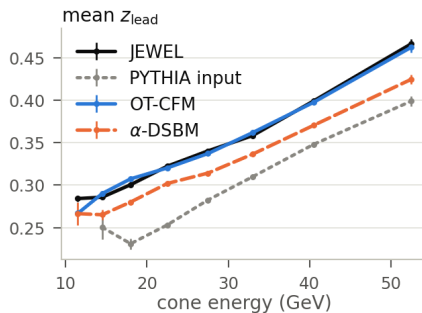
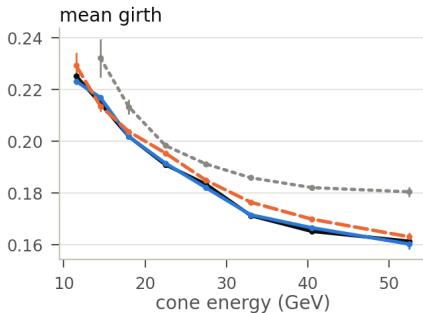


	E	mass	girth	z_{lead}
JEWEL vs JEWEL	0.011	0.014	0.009	0.019
PYTHIA input	0.55	0.88	0.10	0.23
— OT-CFM	0.025	0.021	0.012	0.008
- - α -DSBM	0.26	0.33	0.06	0.10

Distance to 20k held-out JEWEL jets ($W1 / \sigma$); lower is better.

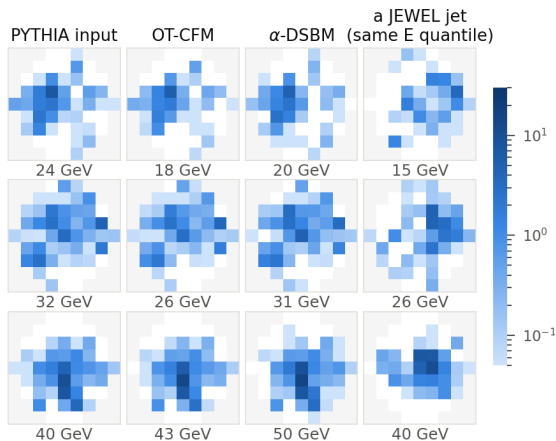
- **OT-CFM** is as close to JEWEL as a second JEWEL sample, within statistical errors; the same holds for p_T^D , z_g , R_g and for jets re-found with anti- k_T
- **α -DSBM** closes 51-64% of the gap

Jet shapes at fixed energy are reproduced too



- at the same energy, JEWEL jets are **narrower** and **harder** (higher z_{lead}) than PYTHIA jets
- **OT-CFM** follows both trends in every energy bin; α -**DSBM** gets part of the way

Each output keeps its input jet

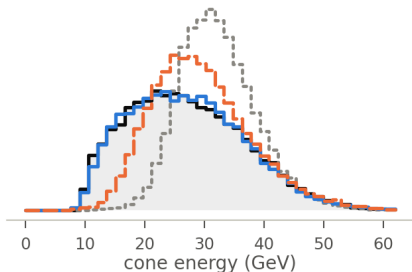


- **OT-CFM**: input-output correlation 0.94 (energy), 0.99 (girth), 0.91 (mass)
- every output is closer in shape to its own input than to another, random input jet, by $11\times$ on average
- a random JEWEL jet has the right distributions but correlation 0: matching distributions alone proves nothing
- **α -DSBM**: 0.80 (energy), 0.98 (girth); two samples of one jet differ more than a sample differs from its input

Fixed held-out inputs (cone-energy quantiles 0.1, 0.5, 0.9). The JEWEL jet is a reference at the same energy quantile, not a truth.

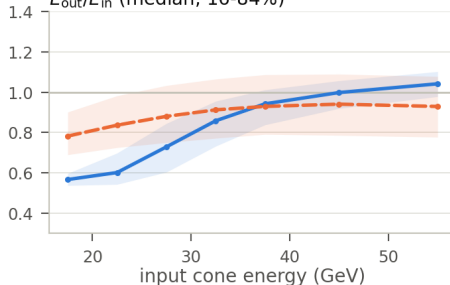
Caveat: the energy change mirrors the two samples

energy spectra



— JEWEL - - - PYTHIA input — OT-CFM - - - α -DSBM

$E_{\text{out}}/E_{\text{in}}$ (median, 16-84%)



- the PYTHIA sample starts near 20 GeV (presumably its generator-level jet requirement); JEWEL reaches down to 10 GeV
- **OT-CFM** lowers 20-25 GeV jets by 40% and leaves jets above 40 GeV unchanged: a map between two differently selected samples, not a quenching mechanism

Summary

Shown

- trained without pairs in 2 GPU hours, **OT-CFM** produces jets statistically indistinguishable from held-out JEWEL: 7 observables, their energy dependence, re-found jets
- each output stays tied to its input (correlations 0.91-0.99)
- α -**DSBM**, same budget: about half way

Not shown

- a per-jet medium modification: there is no per-jet truth, and optimal transport picks the least-change map
- physics in the energy shift: it reflects how the samples were selected
- robustness: one seed

Next: train two more seeds. Does the same PYTHIA jet get the same output, not only the same distributions?

Practical: the ODE needs 128 network calls (a 4-step solve barely moves the jets); towers that should be empty get ~ 0.002 GeV, 0.1% of the jet energy.